ACHRAF RAHIBE

UAV & AVIONICS SYSTEMS ENGINEER

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UAV and avionics engineer developing aerial, ground and maritime systems from architecture and integration through testing. Leads multidisciplinary electronics work and delivers autopilot configuration, diagnostics and technical documentation.

TECHNICAL SKILLS

Flight Control & Avionics
ArduPilot, PX4, INAV, Betaflight; Mission Planner, QGroundControl, Gazebo/SITL; MAVLink, PID/FFT.

Embedded & Interfaces
C/C++, Python, VHDL, Linux, Git; STM32/ESP32, Jetson/Raspberry Pi; CAN/DroneCAN, UART, SPI, I2C.

Vision & Navigation
YOLOv8/ByteTrack, RTSP/GStreamer; A8C gimbal, GNSS/RTK, telemetry, AIS and ADS-B.

Electronics & Prototyping
ESC/VESC, BMS and power interfaces; KiCad/Altium familiarity; FDM 3D printing.

CAD & Simulation
Onshape, AutoCAD Electrical, Gmsh; CNC files, BOMs and assembly planning.

Design & Communication
Photoshop, Illustrator; graphic design, consultations and technical coordination.

EDUCATION

State Engineer Diploma
Electrical Engineering and Telecommunications; Industrial Control. FSTM, Hassan II University.
2021-2025 | Bac+5

DUT in Electrical Engineering
Electronics and Industrial Computing. EST Fès, Université Sidi Mohamed Ben Abdellah.
2019-2021

Technical Baccalaureate
Electrical Sciences and Technology. Lycée Technique Anoual, Casablanca.
2018-2019

LEADERSHIP

ROBOTHINGS: Founder/President (2023-2025).
ENACTUS FSTM: Marketing & Graphic Design Lead (2022-2023).

LANGUAGES

Arabic: Native
English & French: Fluent
German: Basic

HOBBIES

Surfing, reading, graphic design.

PROFESSIONAL EXPERIENCE

FARASHA SYSTEMS | Rabat, Morocco

Electronics Team Manager — Mar 2026 - Present

- Lead avionics architecture, component selection and cross-functional integration. Coordinate validation, field troubleshooting and technical delivery across electronics, embedded, RF and mechanical teams.

UAV & Avionics Systems Engineer — Aug 2025 - Mar 2026

- Integrated aerial and maritime platforms, covering autopilots, sensors, payloads and power systems from simulation and bench checks through flight and water trials.

UAV & Avionics Engineering Consultant — 2025 - Present
Upwork, Fiverr and direct clients

- Provide engineering consultations, system integration, autopilot configuration and tuning. Guide field teams through live calls, log-based troubleshooting, corrective actions and mission preparation.

CAPGEMINI ENGINEERING | Casablanca

Technical Documentation Technician / Consultant — Jul - Sep 2023

- Produced Stellantis after-sales diagnostic and service documentation, following a four-month internship studying automatic emergency braking.

ENGINEERING PROJECTS

Fixed-Wing UAVs
ArduPlane / PX4 ▪ Gazebo / SITL ▪ DJI O4 Pro

- Designed V-tail, pusher and twin-motor conventional-tail aircraft, including differential thrust. Took airframe and avionics design through SITL mission testing, real flights, mission planning and FPV camera/goggle integration.

VTOL & QuadPlane Systems
Cube / Pixhawk ▪ ArduPilot / PX4 ▪ 3D printing

- Integrated flying-wing QuadPlanes and tiltrotors, matching lift/cruise propulsion, avionics and telemetry. Calibrated and tuned hover and transitions, supporting successful client flight trials.

Multirotor & Rotorcraft Development
2-42 inch classes ▪ ArduCopter / INAV / Betaflight ▪ PID / FFT

- Developed FPV, cinewhoop and heavy-lift platforms from 2 to 42 inches, including DARPA Lift Challenge support: propulsion matching, configuration, calibration, troubleshooting, tuning, real missions and helicopter flight-log diagnostics.

Unmanned Surface Vehicles
ArduRover ▪ Waterjets ▪ Jetson Nano ▪ A8C ▪ GNSS / RTK

- Developed a twin-waterjet USV from architecture to water trials, combining differential thrust, autonomous navigation, FPV and AI detection. Integrated propulsion, power and safety interfaces; supported AIS on related vessel work.

Integrated UAV-Rover System
ArduPilot / ArduRover ▪ CUAV X7 ▪ VESC ▪ MAVLink

- Developed and tested the complete drone-rover system from scratch: aircraft and ground autopilots, motor control, communications and docking, with synchronized actuators, coordinated control and failsafe checks.

Embedded Vision & Avionics
Jetson / Raspberry Pi ▪ YOLOv8 / ByteTrack ▪ RTSP

- Integrated camera, tracking and video interfaces; reviewed companion-computer baseboards. Prepared AprilTag landing architecture, MAVLink interfaces and staged verification criteria.